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3D Vision Language Models

Computational Cognition, Vision, and Learning

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Overview

- We present recent works on 3D Vision Language Models (3D VLMs), focusing on identifying the limitations of current VLMs for tasks involving 3D spatial reasoning and developing alternatives.
- **The first part** introduces a series of our studies on controlled realistic-synthetic data. We develop 3D VLMs, based on robust visual modules, which significantly outperform open source and proprietary VLMs.
- **The second part** extend our approach to real world images to 3D VLMs which significantly outperform existing VLMs on questions involving 3D reasoning.

Background

- Vision Language Models (VLMs) are very successful for reasoning and answering questions about images. They can be applied to almost all visual tasks and applications, including those important to the mission of the Army.
- **But VLMs have limitations. They are reported to underperform on questions about images which involve spatial relations and 3D knowledge, including physical reasoning.**
- We conjecture that these limitations are because their training -- on huge datasets of images and images paired with text – **does not involve 3D groundtruth**. Previous studies by CCVL, inspired by Cognitive Science, have argued that vision models should have 3D world knowledge.

3D VLMs

- 3D VLMs have the following properties.
- (1) **They are 3D-aware**: meaning that they have been trained on images with 3D groundtruth, hence enabling them to outperform conventional VLMs on tasks involving 3D world knowledge.
- (2) **They are 3-D explicit**: meaning that they involve algorithms which can estimate 3D representations of the images explicitly and use it for diagnosing the failure models of 3D-VLMs.
- *This talk is based on four recent works: (1) Xingrui Wang et al. ICLR 2025, (2) Xingrui Wang et al. CVPR 2025 highlight, (3) Wufei Ma et al. CVPR 2025 highlight, (4) Wufei Ma et al. In submission 2025).*

Our work involves three datasets

- (1) **SuperClevr**: A series of datasets of realistic-synthetic images and videos with complete 3D groundtruth and questions (2023-2025). They enable us to systematically study limitations of VLMs. They are used for **training and testing** 3D-VLMs.
- (2) **3DSRBench (Wufei Ma et al. 2025)**: A large dataset of real-world images annotated with partial 3D groundtruth. This are used for **testing** 3D-VLMs.
- (3) **3D-Informed Data**. A Dataset of real-world images with 3D groundtruth. This is constructed by: (i) Combining existing datasets of images with 3D groundtruth. (ii) Creating new datasets with 3D pseudo-label groundtruth, using foundational models validated by annotators. These are **only used for training** 3D-VLMs.

Part 1: VLMs and 3D VLMs on SuperClevr

- SuperClevr datasets were created so that Vision-Language algorithms could be studied systematically under controlled settings and with 3D groundtruth.

The SuperClevr datasets

SuperCLEVR

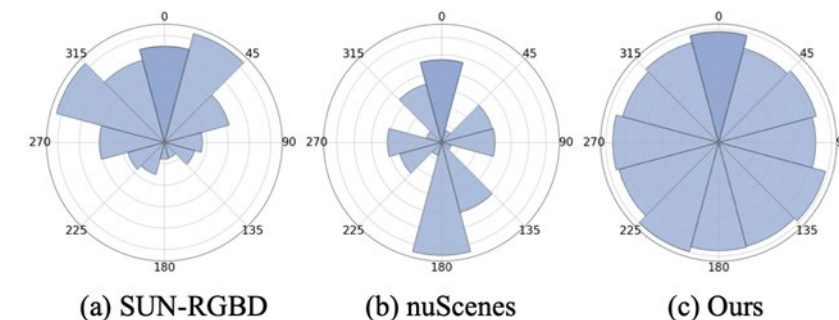
SuperCLEVR-3D
NeurIPS 2023

SuperCLEVR-Physics
ICLR 2025

Spatial457
CVPR 2025 (highlight)

1. Controlled generation
2. **Complete** 3D ground-truths (pose, segmentation, visibility, etc.)
3. Useful for studying failures of vision module and language reasoning

Note: 2025 SuperClevr images are more realistic.



SuperClevr3D History: the 2023 versions.

- The first dataset with 3D groundtruth for evaluating Vision-Language models. (X. Wang et al. NeurIPS 2023) based on (Z. Li et al. CVPR 2023. Oral).
- This dataset enables questions about parts, 3D poses, and occlusions. Note: these synthetic images lacked realism, improved by 2025.

Questions about parts, poses, and occlusions.



Part

- Q: What is the name of **the brown part of** the large rubber thing? *Wheel*
Q: What is the material of the trunk that **belongs to the same object as** the purple part? *Metallic*

3D Pose

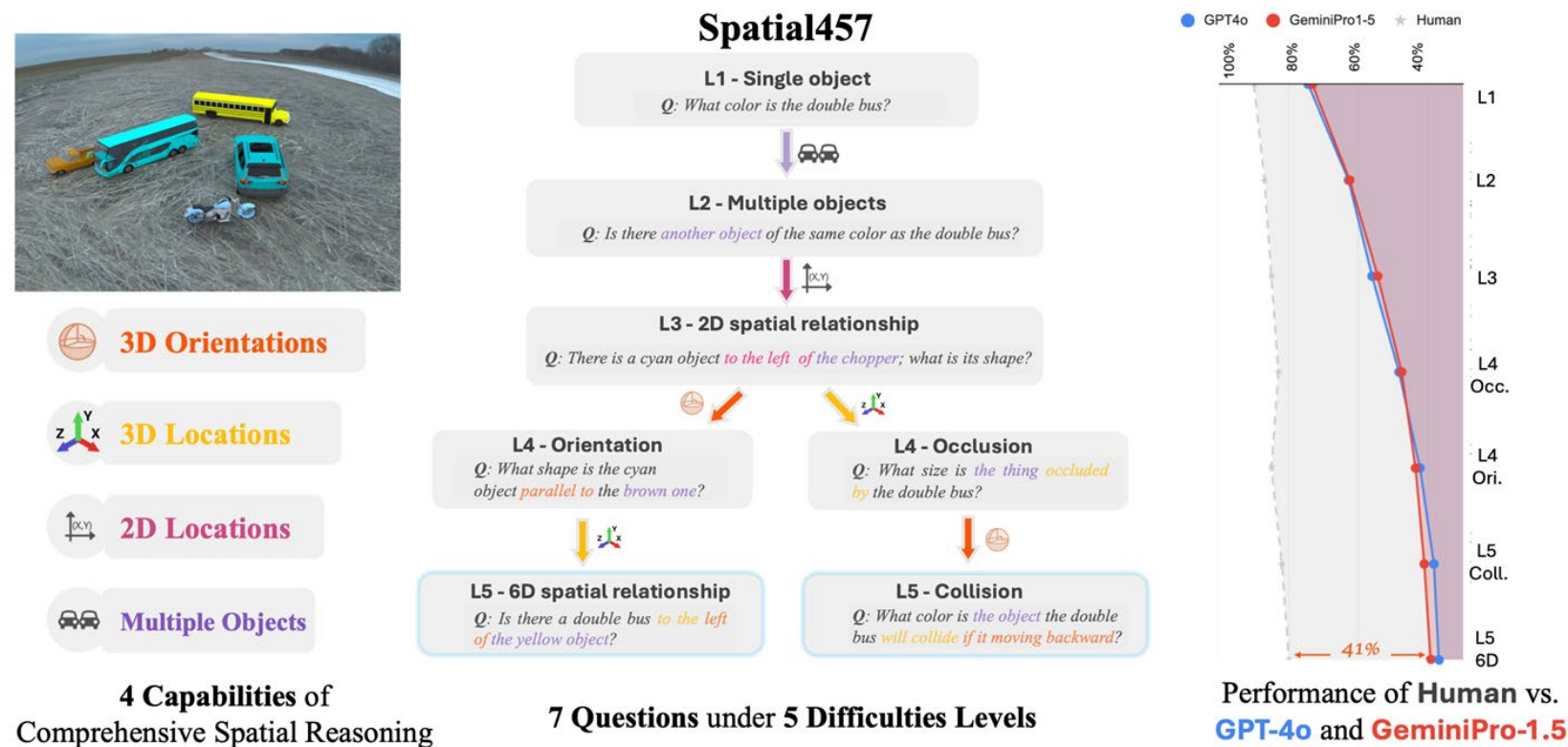
- Q: Which direction the double bus **is facing?** *Left*
Q: What is the color of the small object which **faces to the right?** *Truck*

Occlusion

- Q: Is the bumper of the purple SUV **occluded?** *No*
Q: What is the size of the aeroplane whose wing is **occluded?** *Small*

Comprehensive Spatial Reasoning (CVPR 2025).

- We improved the realism of SuperClevr and evaluated the performance of VLMs and Humans on Comprehensive Spatial Reasoning Tasks. There are 5 Levels of difficulty. VLM performance -- GPT-4 and GeminiPro-1.5 -- degrades rapidly from L1 to L5.



Xingrui Wang et al. CVPR 2025: PulseCheck457: A Diagnostic Benchmark for Comprehensive Spatial Reasoning of Large Multimodal Models. CVPR 2025.

Studying 3D Spatial Reasoning in Simulation

Questions involving 3D spatial relationships.



Q: How many yellow objects are **to the left** of the red wagon?

A: 1

(a.1) 2D spatial



Q: How many yellow objects are **on the left hand side** of the red wagon?

A: 3

(a.2) 6D spatial



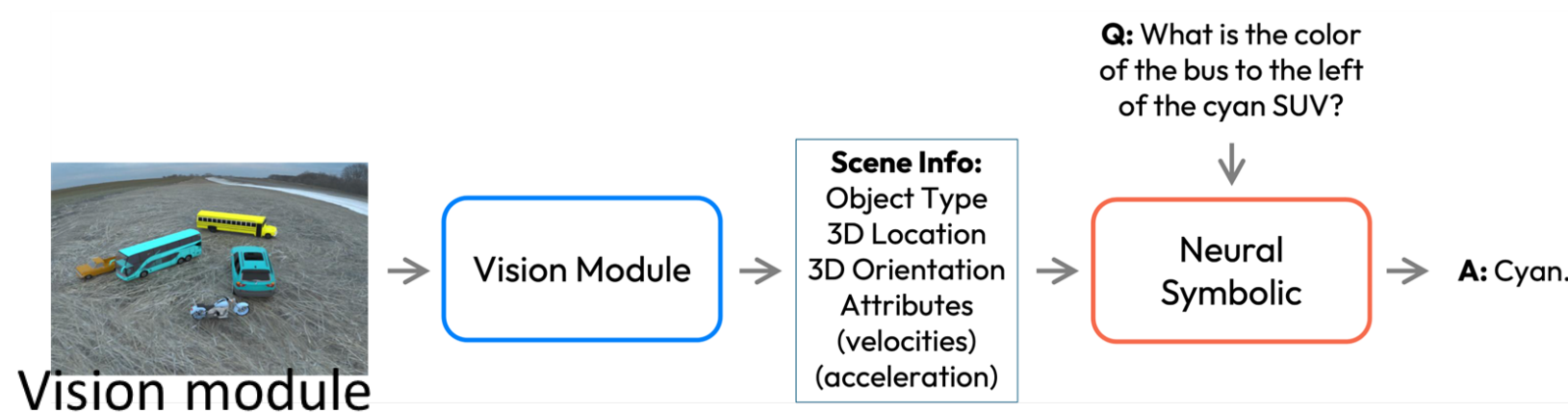
Q: Is the red wagon will **collide with** a yellow regular bus **if it moving forward**?

A: No

(a.3) Collision

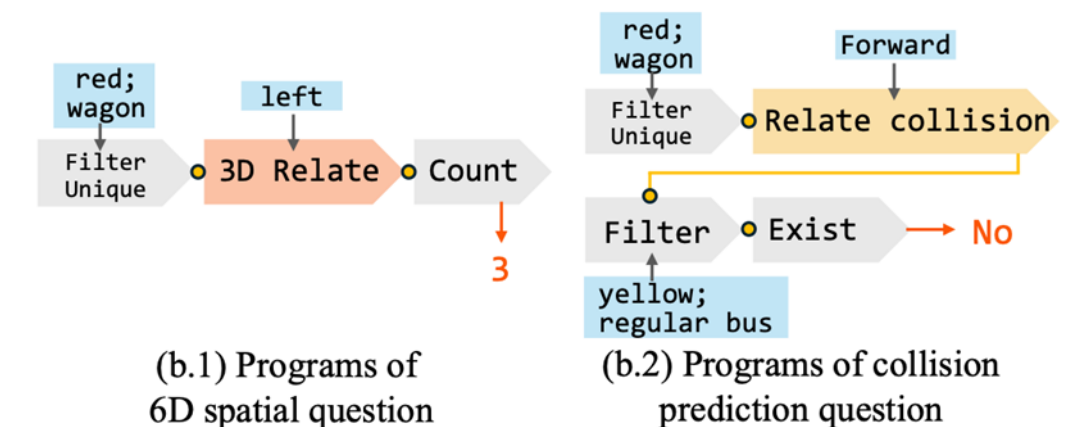
3D VLMs with 3D Image Parsing.

- We develop a 3D VLM – **PO3D-VQA** -- which parse the image and output 3D representations of the objects (their shape, their 3D pose, their parts). This is based on previous work by our group on real world images (e.g., A. Jesslen et al. CVPR 2024). This 3D VLM answers questions about the image based on these representations.



Vision module

- CNN: object color, material, etc.
- NeMo6D [1]: object type, 3D location, 3D orientation
- NS-4Dynamics [2]: velocity and acceleration by integrating physical priors



Summary Results: X. Wang et al. CVPR 2025.

The 3D VLMs, using 3D Image Parser based on neural symbolic methods, significantly outperform previous open-sourced and proprietary VLMs.

3D VLM performance approaches human level.

Difficulty Level	Level 1	Level 2	Level 3	Level 4		Level 5	
Questions type	Single Object	Multi-Obj.	2D Spatial	Occlusion	3D Pose	Collisions	6D Spatial
Random	33.05	32.77	33.47	22.04	18.99	21.02	19.41
GPT-4o [29]	74.46	62.88	56.14	48.40	42.41	38.41	37.01
GeminiPro 1.5 [37]	73.26	62.54	54.49	47.65	43.67	41.19	39.36
Claude 3.5 Sonnet [3]	68.24	57.40	54.19	30.84	38.40	35.34	33.48
Qwen2-VL-7B-Inst. [39]	71.96	61.44	55.34	27.87	34.29	36.58	33.75
InternVL2-8B [38]	58.11	58.76	57.40	33.25	32.32	34.30	34.30
LLaVA-v1.5-7B [26]	44.87	44.72	42.20	24.34	24.55	23.63	23.86
LLaVA-NeXT-vicuna-7B [27]	50.72	49.47	46.01	29.68	29.35	31.95	31.95
LLaVA-NeXT-llama3-8B [41]	52.15	49.73	45.92	30.31	29.77	32.12	32.12
PO3D-VQA [45]	86.46	82.55	80.64	70.49	81.40	68.12	71.06
Human	89.97	86.83	84.95	82.76	84.95	81.82	79.94

SuperClevr for Physics: X. Wang et al. ICLR 2025.

- We extend SuperClevr to videos using physics simulators. This enables us to test questions that involve physical understanding, can make predictions, and perform counterfactual reasoning.
- We extend our 3D-VLM to videos – **NS-4DPhysics** -- using Bayes-Kalman filters so it integrates information over time. X. Wang et al. ICLR. 2025. Beats alternatives by 20% AP (average).

Input Videos			Questions	Answers	
(a)				Factual Question 1 Is the suv moving faster than the yellow bus?	No
				Factual Question 2 What is the color of the bus which collides with the suv?	Green
(b)				Predictive Question Will the red bus collide with the yellow object?	Yes
(c)				Counterfactual Question Would the bus collide with the minivan if it were standing still from the beginning?	No

Experimental Results

- We find that 3D VLMs based on Image Parsing with Physical Knowledge (NS-4DPhysics) outperform VLMs for factual (velocity, acceleration, collision), prediction, and counterfactual reasoning.

	Average	All	Factual			Predictive	Counterfactual
			Vel.	Acc.	Col.		
CNN+LSTM	48.03	40.63	41.71	56.79	25.37	56.04	47.42
FiLM (Perez et al., 2018)	50.18	44.07	48.58	53.09	26.87	54.94	51.54
NS-DR (Yi et al., 2019)	51.44	51.44	55.63	46.34	46.86	-	-
PO3D-VQA (Wang et al., 2024)	<u>62.93</u>	<u>61.22</u>	<u>62.21</u>	<u>73.17</u>	<u>51.20</u>	65.33	62.24
InternVideo (Wang et al., 2022)	52.62	51.07	59.29	49.08	36.06	54.74	59.18
Video-LLaVA [†] (Lin et al., 2023)	38.09	37.04	37.62	52.76	23.56	38.78	40.88
PLLaVA [†] (Xu et al., 2024)	59.24	54.61	55.00	63.80	46.63	<u>67.52</u>	<u>73.47</u>
GPT-4o [†]	51.59	50.82	51.19	57.67	44.71	54.38	50.00
GPT-4o + reasoning [†]	56.06	55.50	58.81	57.67	47.12	56.93	58.16
NS-4DPhysics	82.64	87.70	88.66	83.73	88.46	85.71	74.51

Summary of Part 1.

- These two studies on realistic-synthetic data show that **3D VLMs with 3D Image Parsing can significantly outperform conventional VLMs** when evaluated on these controlled datasets of realistic-synthetic data with complete 3D groundtruth. This approach is being scaled up, **but at present is limited to a small number of objects in simple realistic-synthetic images.**
- This motivates us to develop a parallel approach which takes advantage of the strengths of conventional VLMs – **their vast implicit knowledge of the world and their ability to perform many types of reasoning and answer a large variety of questions.** We develop 3D VLMs that can perform 3D spatial reasoning on **real world images** better than conventional VLMs, **while maintaining high performance on answering other types of questions.**

Part 2: 3D-VLMs on Real Images

We take existing VLMs and make them 3D-aware to improve its ability for answering questions involving 3D properties. How to improve the 3D awareness of the model and the 3D reasoning abilities?

1. We need dataset of real images to test 3D properties.
2. We need data for to train for 3D-awareness.
3. We need to fine-tune VLMs to make them 3D-aware and ideally 3D-explicit.

For **testing**, we construct a new dataset annotated with questions that requires understanding properties of the 3D world (but without full 3D annotations).

For **training**, we take existing datasets with 3D groundtruth and create dataset with 3D pseudo-groundtruth from foundation models (e.g., depth anything) checked by annotators.

In both cases, **we design questions and answers for 3D properties.**

(1) Dataset for Testing 3D Properties

- **For testing:** We created 3DSRBench. A large dataset of real-world images which we annotated using partial 3D groundtruth. The questions require understanding the 3D world.



Distance Relationships

The pedestrian is closer to the viewer than the cyclist.

Orientation Relationships

White van is facing the pedestrian instead of the cyclist.

Spatial Reasoning Relationships

From the perspective of the pedestrian, the white van is on the right side of the pedestrian.



A subset, **SpatialVQA**, of this dataset was available when our 3D VLMs were trained. The full dataset is in submission (Wufei Ma et al. 3DSRBench. 2025).



3D-Informed Data

1M 3D-Informed Probing Data description on depth, distance, & orientation

1M 3D-Informed Instruction Tuning Data conversations about 3D spatial relationships

3D-Informed Architecture

Mixed encoders, 3D-aware visual encoders, etc.

3D-Informed Training

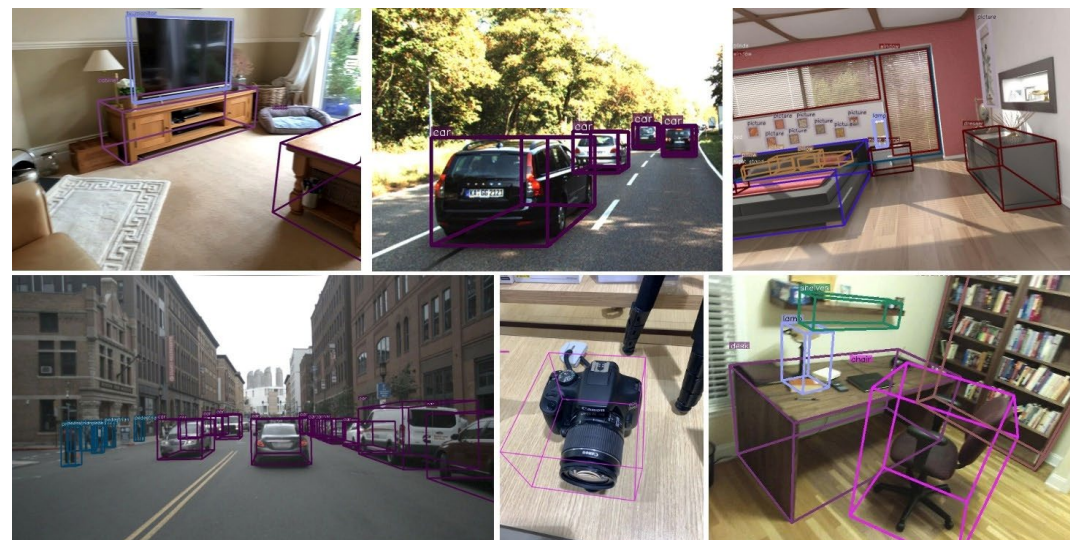
Pre-pretraining, alignment & finetuning

Images with Real annotations

- ## Images with Pseudo annotations

[illegible]

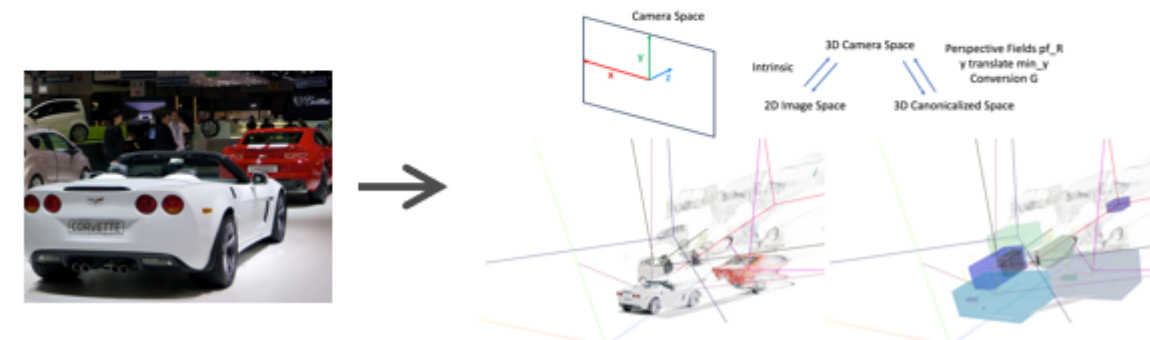
ImageNet3D+ with 3D poses



Omni3D with 3D locations

Unlabeled
Internet Image

3D Ground-truths
pseudo annotated by
Foundation models.



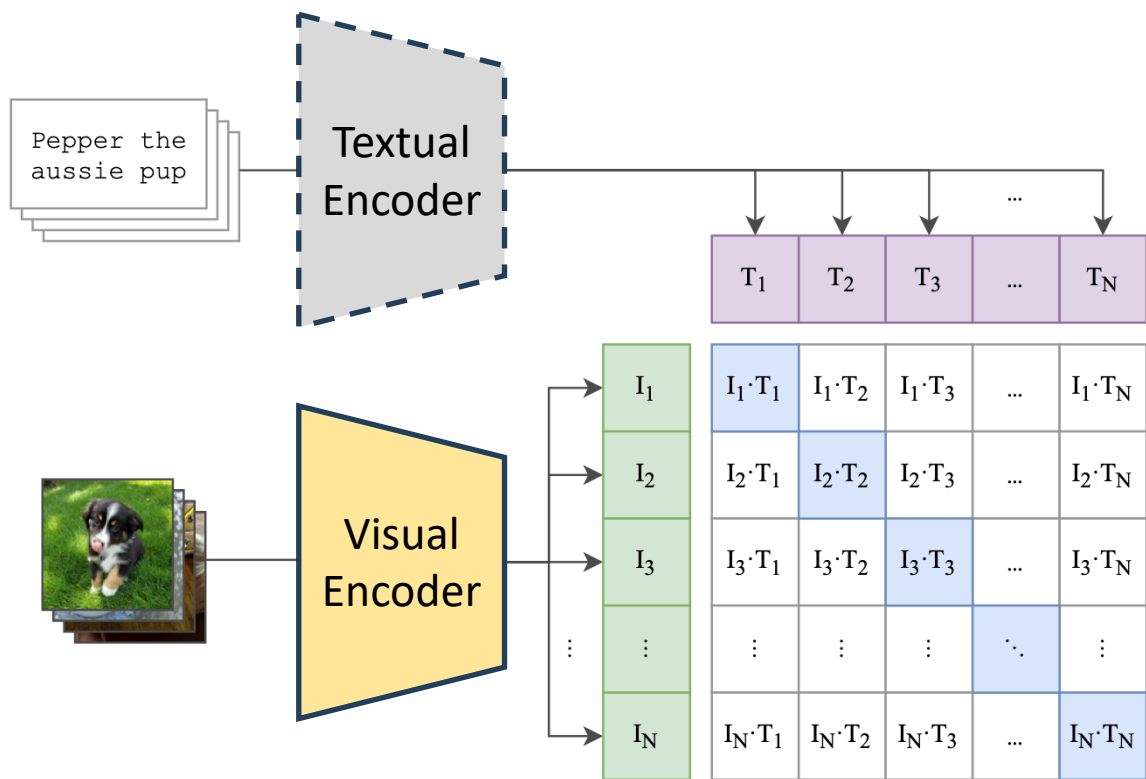
3D VTM: based on LLaVA-v1.5.

- This is our first project. (Wufei Ma et al. CVPR 2025. Highlight).
- We use LLaVA-v1.5 as a base VTM.
- We finetune LLaVA-v1.5 on the 3D-Informed Data.
- LLaVA-v1.5 has three components:
 - (1) A visual encoder. (2) A connector. (3) A Large Language Model.
- Details of the finetuning follow.

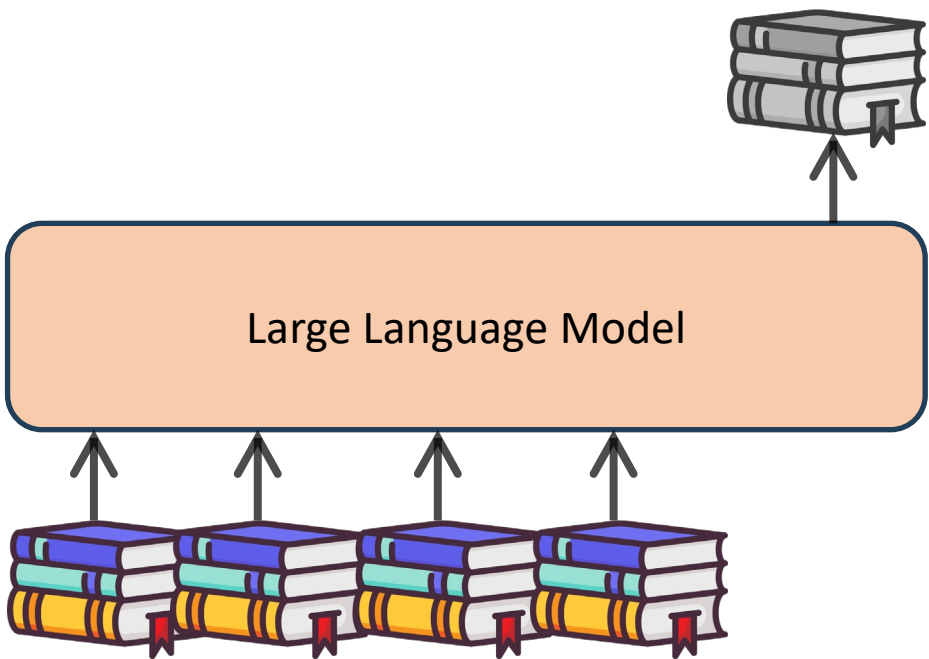
How is LLaVA-v1.5 trained?

Stage 0: Pre-train visual encoder and large language model separately

Contrastive Language-Image Pre-training



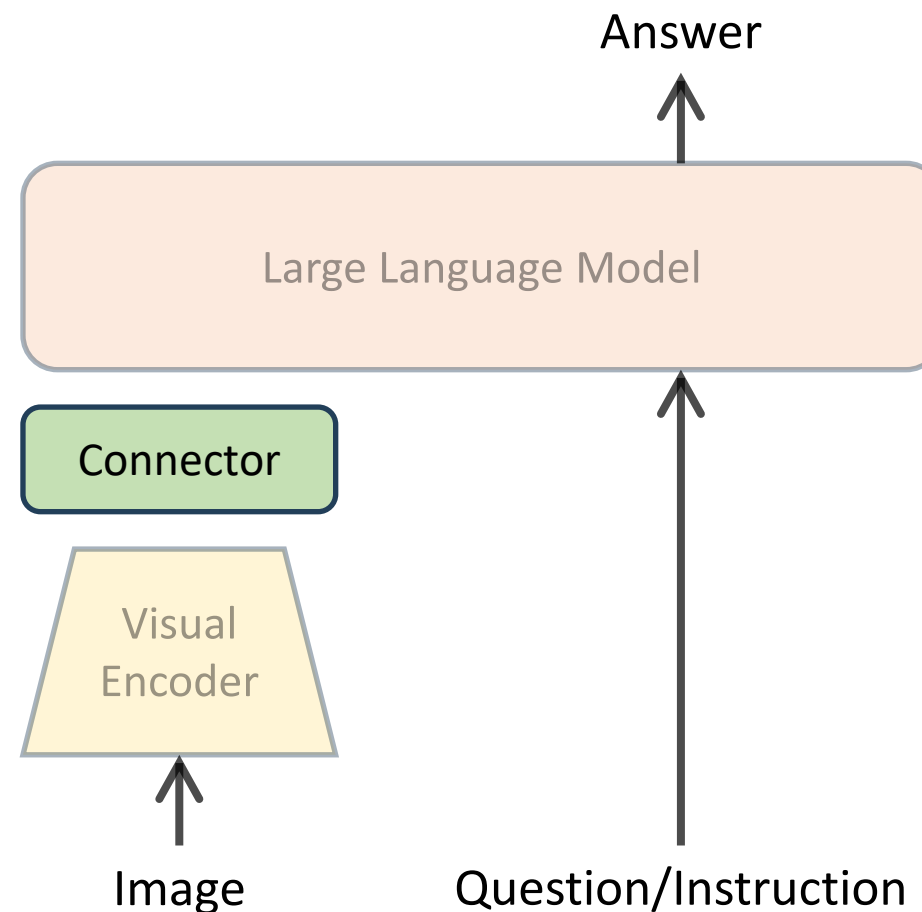
Causal Language Modeling



How is LLaVA-v1.5 trained?

Stage 0: Pre-train visual encoder and large language model separately

Stage 1 (Multimodal alignment): train the visual connector while freezing the visual encoder and large language model

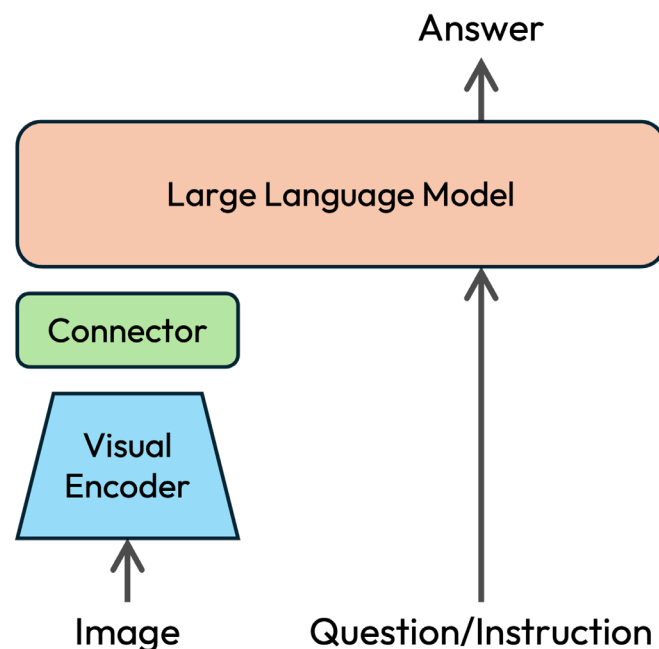


How is LLaVA-v1.5 trained?

Stage 0: Pre-train visual encoder and large language model separately

Stage 1 (Multimodal alignment): train the visual connector while freezing the visual encoder and large language model

Stage 2 (Visual instruction tuning): finetune the full model jointly on high-quality QA or conversation data



Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?

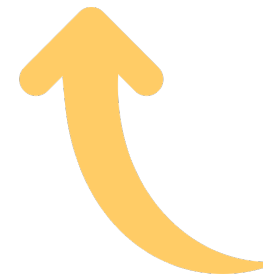
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

How is LLaVA-v1.5 trained?

Stage 0: Pre-train visual encoder and large language model separately

Stage 1 (Multimodal alignment): train the visual connector while freezing the visual encoder and large language model

Stage 2 (Visual instruction tuning): finetune the full model jointly on high-quality QA or conversation data



How to inject 3D awareness and reasoning abilities?

- Depth
- Height
- Location
- Pose/orientation
- Spatial relationship

=> We create 3D-informed data and adopt 3D-aware training for the vision language model.

3D-Informed Training: Fine-Tuning

Stage 0

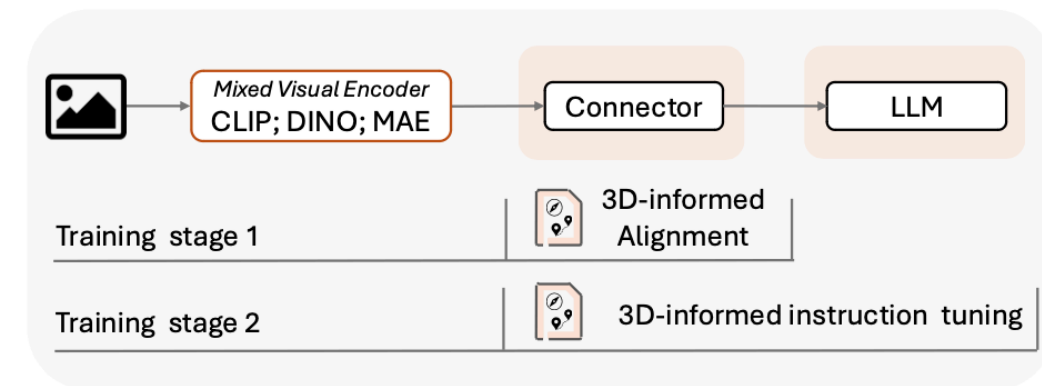
1. Mixed encoder: CLIP, DINOv2, etc.
2. 3D-informed encoder: CLIP + 3D finetuning

Stage 1 (multimodal alignment)

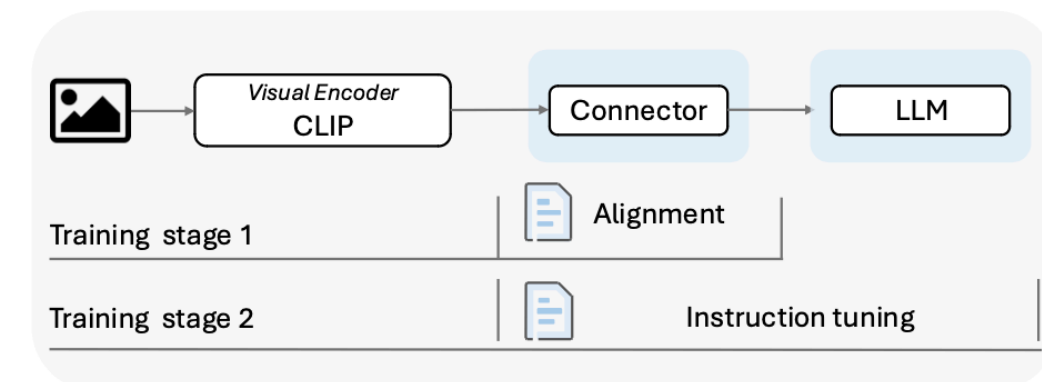
1. 3D-informed alignment

Stage 2 (visual instruction tuning)

1. Mixing 3D-informed and LLaVA datasets

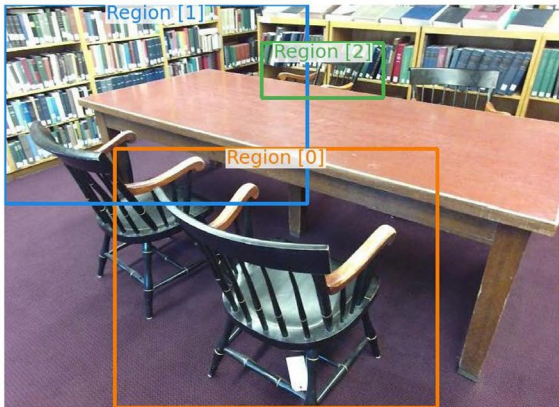


(a) our proposed design



(b) design of LLaVA-v1.5 (CVPR'24)

SpatialVQA Dataset: Testing 3D properties.



Distance Relationships
✓ 3D Location ∅ 3D Orientation

User

Which object is closer to the chair in Region [0], the bookcase in Region [1] or the chair in Region [2]?

3DI-LMM

The chair in Region [2] is closer to the chair in Region [0].



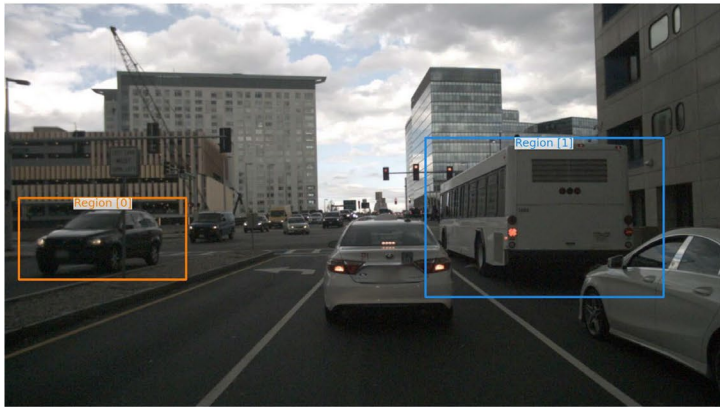
Orientation Spatial Relationships
✓ 3D Location ✓ 3D Orientation

User

Are the car in Region [0] and the car in Region [1] facing the same direction, or different directions?

3DI-LMM

The car in Region [0] and the car in Region [1] are facing different directions.



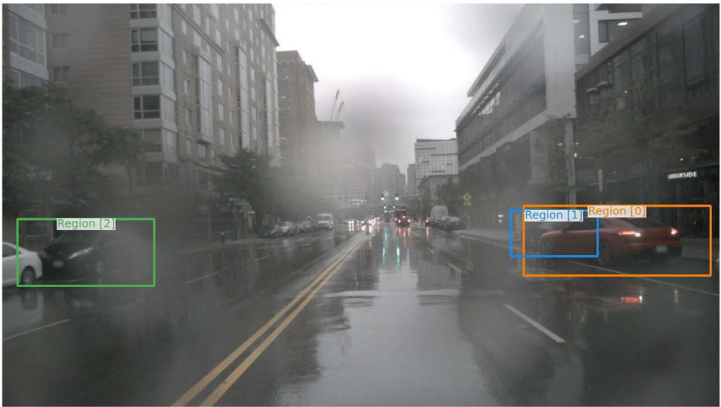
Spatial Reasoning Relationships
✓ 3D Location ✓ 3D Orientation

User

If I stand at the location of the car in Region [0] facing where it is facing, where is the bus in Region [1]?

3DI-LMM

The bus in Region [1] is on the left side of the car in Region [0].



Spatial Reasoning Relationships
✓ 3D Location ✓ 3D Orientation

User

Is the car in Region [0] oriented towards the car in Region [1] or the car in Region [2]?

3DI-LMM

The car in Region [0] is facing towards the car in Region [1].

Results

1. Our SpatialLLM significantly outperforms all previous open-sourced and proprietary models.

Model	Avg.	3D Dist.	3D Orient.	3D Spatial Rel.
<i>Proprietary</i>				
GPT-4o [1]	54.0	59.4	52.4	50.8
GPT-4o Mini [1]	45.4	46.9	39.3	48.5
Claude 3.5 Sonnet [4]	49.3	50.8	44.5	51.5
<i>Open-sourced</i>				
LLaVA-v1.5-7B [39]	47.7	50.6	46.4	46.2
LLaVA-NeXT-8B [40]	50.6	63.5	49.2	41.5
Cambrian-8B [58]	51.7	59.8	46.4	48.9
<i>Spatial-centric</i>				
SpatialVLM-13B [14]	52.2	64.3	48.5	45.8
Ours	62.7	86.3	52.9	50.8

Results

- 1. Our SpatialLLM significantly outperforms all previous open-sourced and proprietary models.
- 2. Best recipe is a combination of 3D-informed multimodal alignment and 3D-informed visual instruction tuning => 3D awareness is important!

<i>3D-informed Tuning for Stage 2</i>								
CLIP [48]+DINOv2 [47]	Llama3-8B [18]	–	CC558K [39]	Mix665K [39] + 3DI-Ft1M	59.7	85.2	52.1	44.8
<i>3D-informed Alignment for Stage 1</i>								
CLIP [48]+DINOv2 [47]	Llama3-8B [18]	–	CC558K [39] + 3DI-Ft1M	Mix665K [39] + 3DI-Ft1M	57.5	80.8	46.9	46.4
CLIP [48]+DINOv2 [47]	Llama3-8B [18]	–	CC558K [39] + 3DI-Pb-OI1M	Mix665K [39] + 3DI-Ft1M	60.0	82.1	46.9	51.6
CLIP [48]+DINOv2 [47]	Llama3-8B [18]	–	CC558K [39] + 3DI-Pb-IN166K	Mix665K [39] + 3DI-Ft1M	62.7	86.3	52.9	51.0

Results

- 1. Our SpatialLLM significantly outperforms all previous open-sourced and proprietary models.
- 2. Best recipe is a combination of 3D-informed multimodal alignment and 3D-informed visual instruction tuning.
- 3. Finetuning CLIP with 3D-informed data hurts final performance. (Trade-off between 3D awareness and generalization abilities.)

CLIP [48]+DINOv2 [47]	Llama3-8B [18]	–	CC558K [39] + 3DI-Pb-IN166K	Mix665K [39] + 3DI-Ft1M	62.7	86.3	52.9	51.0
3D-informed pretraining for vision encoder in stage 0								
CLIP [48]+DINOv2 [47]	Llama3-8B [18]	3DI-Pb-OI1M (CLIP [25])	CC558K [39] + 3DI-Pb-IN166K	Mix665K [39] + 3DI-Ft1M	60.3	85.0	50.5	47.5
CLIP [48]+DINOv2 [47]	Llama3-8B [18]	3DI-Pb-OI1M (NTP [34])	CC558K [39] + 3DI-Pb-IN166K	Mix665K [39] + 3DI-Ft1M	62.6	85.2	52.9	51.5

Limitations

No explicit 3D representations or reasoning process

Limited reasoning abilities

No interpretability

Cannot analyze the failure modes: vision or language
module

In our second 3D-VLM paper, we study:

1. How to answer questions with explicit 3D representations and chain-of-thought reasoning.
2. Study the failure cases of the 3D spatial reasoning.

LLM Background

- Many recent VLMs and LLMs decompose reasoning and question answering into a sequence of small steps. This breaks down complex questions into smaller, and more tractable, sub-questions.
- This is informally known as **chain of thought reasoning**. It improves performance and **results in better interpretability**.
- It relates to **test-time scaling** methods which achieve good performance by using more computational during inference time.
- This motivates us to use the Qwen-VL model, which is one of the most popular open-weight vision-language model. We finetune it, as before, on our 3D-informed dataset. Chain of thought reasoning encourages the 3D VLM to be both 3D-aware and 3D-explicit.
- **We next show examples of leading proprietary VLMs -- GPT-4.5 and Gemini 2.0 -- illustrating chain of thought, but showing that both lack 3D-awareness (at least on this image).**

GPT-4.5



Question: From the perspective of the man, is the bottled water on the left or right side of him?

Answer: The man seems to be inside a food truck, and from the image, the water bottle is on the left side.

But looking from his perspective: he's facing roughly to the left side of the image (our left).

Since he's holding the foil package in his right hand, the water bottle is probably on his left side.

This makes sense because the foil package is in front of the right side, so the water bottle would logically be opposite, on the **left** side.

Ground-truth: On the **right** side.

A mistake showing that GPT-4.5 reasons correctly but lacks 3D-awareness.

Gemini 2.0 Thinking



Question: From the perspective of the man, is the bottled water on the left or right side of him?

Thinking: Here's my thinking process to determine the relative position of the water bottle:

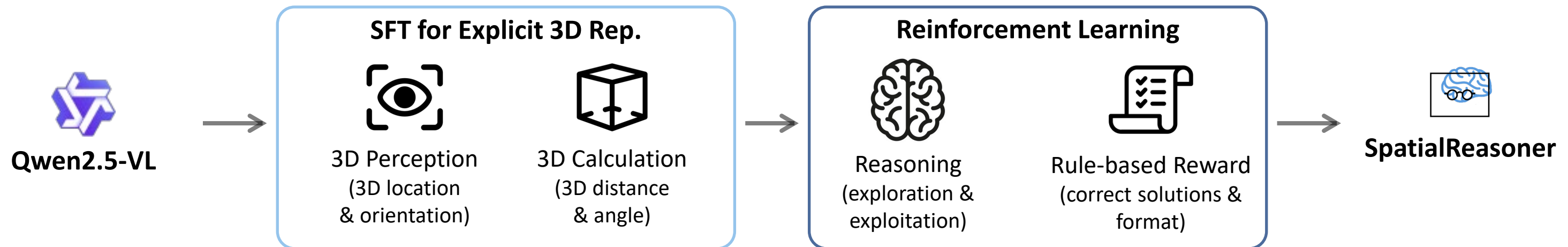
1. ... 2. ...
3. Locate the man in the image: The man is clearly visible in the center of the frame.
4. Locate the bottled water: The bottled water is in the foreground, partially obscuring the man.
5. Determine the man's orientation: The man is facing towards the viewer (or **slightly angled to his right**)...
6. Project the man's perspective: Imagine a line extending straight out from the man's nose or chest. The bottled water is located to the left of this imaginary line, from his viewpoint.
7. ...

Answer: From the perspective of the man, the bottled water is on his **left** side.

Ground-truth: On the **right** side.

This mistake shows Gemini 2.0 also lacks 3D-awareness.

SpatialReasoner: A 3D VLM which is 3D-explicit



New Training data with Explicit 3D Annotations and Chain-of-Thought Reasoning

The 3D location of the bottled water is (0.2, 0.4, 0.6). The 3D location of the person is (0.0, 0.5, 1.0). The vector from the person to the bottled water is hence (0.2, -0.0, -0.4). The left direction of the person is (-0.9, -0.1, 0.5). The cosine similarity between the vector and the left direction is -0.85, corresponding to an angle of 147.89 degrees. The angle is larger than 90 degrees, meaning that the bottled water is on the right of the person. Therefore, the final answer is on the **right**.

Reinforcement Learning for Better Generalization

Optimization Algorithm

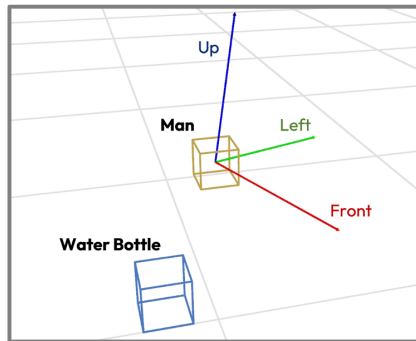
- GRPO
- Format reward
- Process reward

Qualitative Comparison with VTMs



User

From the perspective of the man, is the bottled water on the left or right side of him?



Qwen2.5-VL-72B

✗ 3D thinking ✗ 3D computation

From the perspective of the man, the bottled water is on his **left** side.



ChatGPT o3

✗ 3D thinking ✗ 3D computation

The man seems to be inside a food truck, and from the image, the water bottle is on the left side. But looking from his perspective: he's facing roughly to the left side of the image (our left). Since he's holding the foil package in his right hand, the water bottle is probably on his left side. This makes sense because the foil package is in front of the right side, so the water bottle would logically be opposite, on the **left** side.



Gemini 2.0 Flash Thinking

✓ 3D thinking ✗ 3D computation

(omitted) 3. Locate the man in the image: (omitted) 4. Locate the bottled water: The bottled water is in the foreground, partially obscuring the man. 5. Determine the man's orientation: The man is facing towards the viewer. (omitted) 6. Project the man's perspective: Imagine a line extending straight out from the man's nose or chest. The bottled water is located to the left of this imaginary line, from his viewpoint. (omitted) From the perspective of the man, the bottled water is on his **left** side.



SpatialReasoner

✓ 3D thinking ✓ 3D computation

The 3D location of the bottled water is (0.2, 0.4, 0.6). The 3D location of the person is (0.0, 0.5, 1.0). The vector from the person to the bottled water is hence (0.2, -0.0, -0.4). The left direction of the person is (-0.9, -0.1, 0.5). The cosine similarity between the vector and the left direction is -0.85, corresponding to an angle of 147.89 degrees. The angle is smaller than 90 degrees, meaning that the bottled water is on the right of the person. Therefore, the final answer is on the **right**.

Results: Comparisons with SOTA VLMs

The 3D spatial reasoning abilities of SpatialReasoner outperforms pervious open-sourced and proprietary LVLMs by a wide margin.

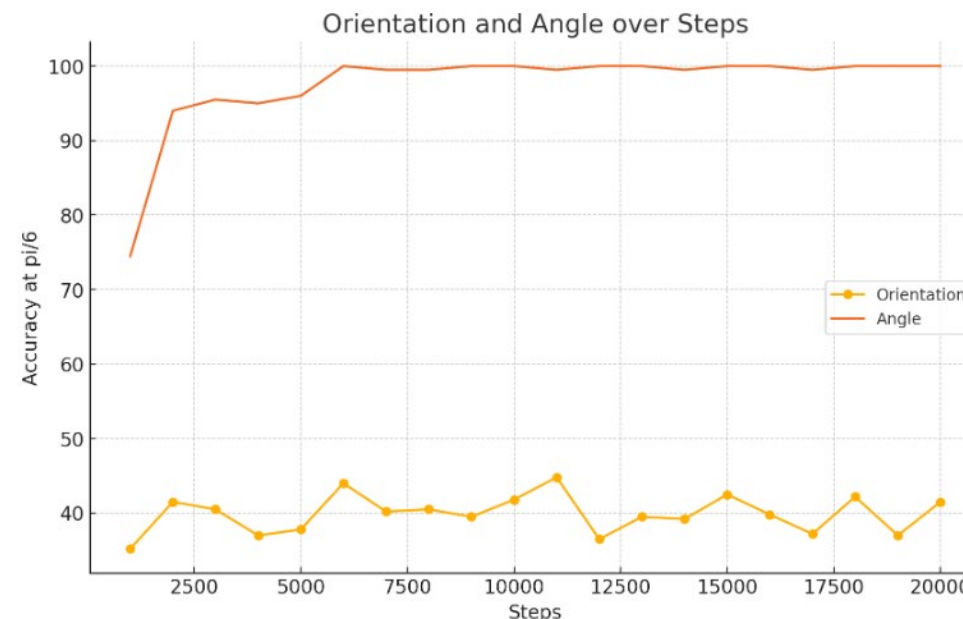
- With explicit 3D representations in reasoning process, we obtain a **4.2%** improvement.

Method	Mean	Height	Location	Orientation	Multi-Object
Open-Sourced Generalist					
LLaVA-v1.5-7B [22]	38.1	39.1	46.9	28.7	34.7
Cambrian-1-8B [38]	42.2	23.2	53.9	35.9	41.9
Qwen2.5-VL-7B-Instruct [3]	48.4	44.1	62.7	40.6	40.5
Open-Sourced Specialist					
SpaceLLaVA [32]	42.0	49.3	54.4	27.6	35.4
SpatialBot [7]	41.0	40.4	54.4	31.9	33.5
SpatialLLM					
Proprietary					
GPT-4o-mini	39.7	44.3	52.4	21.0	36.5
GPT-4o	44.2	53.2	59.6	21.6	39.0
Gemini 2.0 Flash	49.8	49.7	68.9	32.2	41.5
Gemini 2.0 Flash (thinking)	51.1	53.0	67.1	35.8	43.6
Ours					
SpatialReasoner (RL)	54.1	46.4	67.1	49.3	47.1
SpatialReasoner (SFT)	57.0	41.3	72.1	52	51.2
SpatialReasoner (SFT+RL)	58.4	45.8	70.1	55	53.9

Results: Understanding Failures

We also study why our SpatialReasoner still fails to answer some 3D spatial reasoning questions.

1. 3D perception is still challenging -- 3D location and 3D orientation can be inaccurate.
 - Can be improved by adding more 3D-informed data (e.g., synthetic datasets or real datasets with pseudo-annotations).
2. 3D reasoning is accurate and reliable -- angles are computed accurately. Once we obtain correct 3D perception results, we can (most likely) obtain the correct reasoning output.



Future Generalizations: Outdoor Spatial Reasoning

3D Spatial Reasoning



Q: Is the man with a suitcase next to or far from the fire hydrant?

A: They are far from each other.

Collision Reasoning



Q: Will the white car collide with the extended sedan if it reverses by 1 meter?

A: No, there's still plenty of space between them.

Situational Reasoning



Q: Given the current distance between motorcyclist in black and the traffic cone, can a vehicle with a width of 1.8 meters safely pass between them?

A: No.

Conclusion

- 3D knowledge of the world is important for VLMs.
- We present two strategies for creating 3D VLMs which are 3D-aware and 3D-explicit.
- Strategy 1: 3D VLMs based on 3D Image Parsers. (Xingrui Wang et al. ICLR 2025, CVPR 2025 highlight). Very explicit and is currently being scaled up. Significantly outperforms VLMs for spatial and physical reasoning questions and approaches human performance. Currently limited to small realistic-synthetic datasets.
- Strategy 2: 3D VLMs based on existing VLMs finetuned on 3D-informed data. Our first project made 3D-aware VLMs. (Wufei Ma et al. CVPR 2025 highlight). Our second project made them 3D-explicit by using chain of thought (Wufei Ma et al. In submission 2025). These 3D VLMs significantly outperforms VLMs on questions involving spatial reasoning, while maintaining high performance on other types of questions.
- Our strategies are very general and can be generalized to many different tasks.
- Thanks for your attention! Any questions?